Skill - 1

Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

1. Open the browser.
2. Install Wireshark from its official download page.
3. Install it using the default settings.
4. Open Wireshark.

Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as Wi-Fi, Ethernet and Bluetooth.

5. If you are connected through Wi-Fi, select Wi-Fi.
6. Double-click the Wi-Fi interface.
7. Packet capture will start.

Step 3: Generate network traffic

While Wireshark is capturing:

8. Open a new browser tab.
9. Visit a website such as https://www.google.com.
10. Open two or three different websites.
11. Wait for approximately 30 seconds.
12. Return to Wireshark.
13. Click the red Stop button.

You should now see captured packets.

Step 4: Apply display filters

In the Wireshark filter bar, enter each filter separately and press Enter.

Filter	Purpose
tcp	Displays TCP packets.
udp	Displays UDP packets.
icmp	Displays ICMP packets.
tcp.len > 0	Displays TCP packets carrying data.

First, enter: tcp. Observe the packets displayed. Then enter: udp and observe the UDP packets.

The supplied screenshots show tcp, udp, icmpv6 and tcp.len > 0. The ICMPv6 screenshot is recorded as ICMPv6, not as the exact icmp filter.

Step 5: Inspect packet headers

Select a TCP packet from the packet list. In the packet details pane, expand these sections:

14. Frame
15. Ethernet II
16. Internet Protocol Version 4
17. Transmission Control Protocol

Record the following values:

Parameter	Where to find it
Frame length	Frame
Ethernet header	Ethernet II
Source MAC address	Ethernet II
Destination MAC address	Ethernet II
IPv4 header length	Internet Protocol Version 4
IPv4 total length	Internet Protocol Version 4
TCP header length	Transmission Control Protocol
TCP segment / payload length	Transmission Control Protocol

Repeat the process for one UDP packet. Observe the UDP header and its length field.

TCP packet values from the submitted screenshot

Parameter	Observed value
Frame length	490 bytes
Ethernet header length	14 bytes
Source MAC address	20:2b:20:d4:bd:ef
Destination MAC address	4c:d5:87:52:c1:00
IPv4 source address	10.250.10.109
IPv4 destination address	23.65.124.97
IPv4 header length	20 bytes
IPv4 total length	476 bytes
TCP source port	42493
TCP destination port	443
TCP header length	20 bytes (derived)
TCP payload length	436 bytes

UDP packet values from the submitted screenshot

Parameter	Observed value
Frame length	105 bytes
Ethernet header length	14 bytes
IPv4 source address	10.250.10.109
IPv4 destination address	217.79.105.233
IPv4 header length	20 bytes
IPv4 total length	91 bytes
UDP source port	61954
UDP destination port	5001
UDP header length	8 bytes
UDP payload length	63 bytes
UDP length field	71 bytes

Step 6: Calculate header overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Application data = 1500 bytes
- FCS = 4 bytes, if included in the frame representation

Total size including FCS:

14 + 20 + 20 + 1500 + 4 = 1558 bytes

Total overhead:

14 + 20 + 20 + 4 = 58 bytes

Payload efficiency:

(1500 / 1558) × 100 = 96.28%

Note: Wireshark may not capture or display the Ethernet FCS. Use the actual captured packet fields when recording your observations.

Calculation using the selected TCP packet

Frame length = 490 bytes

Total headers = Ethernet header + IPv4 header + TCP header

= 14 + 20 + 20 = 54 bytes

TCP payload = 436 bytes

Payload efficiency = (436 / 490) × 100 = 88.98%

Header overhead percentage = (54 / 490) × 100 = 11.02%

Calculation using the selected UDP packet

Frame length = 105 bytes

Total headers = Ethernet header + IPv4 header + UDP header

= 14 + 20 + 8 = 42 bytes

UDP payload = 63 bytes

Payload efficiency = (63 / 105) × 100 = 60.00%

Header overhead percentage = (42 / 105) × 100 = 40.00%

Step 7: Fill in your observation table

Record values from your own Wireshark capture.

Parameter	TCP Packet	UDP Packet
Frame length	490 bytes	105 bytes
Ethernet header length	14 bytes	14 bytes
IP header length	20 bytes	20 bytes
Transport header length	20 bytes	8 bytes

IP total length	476 bytes	91 bytes
Payload length	436 bytes	63 bytes
Total header overhead	54 bytes	42 bytes
Payload efficiency	88.98%	60.00%

Answer these questions

1. What is encapsulation?
Encapsulation is the process in which each network layer adds its own header to the data received from the layer above it before transmission.

2. What is the purpose of the Ethernet header?
The Ethernet header contains the source and destination MAC addresses and identifies the upper-layer protocol. It helps deliver the frame across the local network.

3. What information is stored in the IP header?
The IP header contains the source and destination IP addresses, packet length, version and other information required to forward the packet across networks.

4. What is the difference between TCP and UDP headers?
TCP has a larger, variable-length header and supports reliable, ordered delivery using sequence and acknowledgement information. UDP has a simple 8-byte header containing source port, destination port, length and checksum.

5. Why does each protocol add a header?
Each protocol adds control information required to perform its responsibility, such as local delivery, routing or transport between applications.

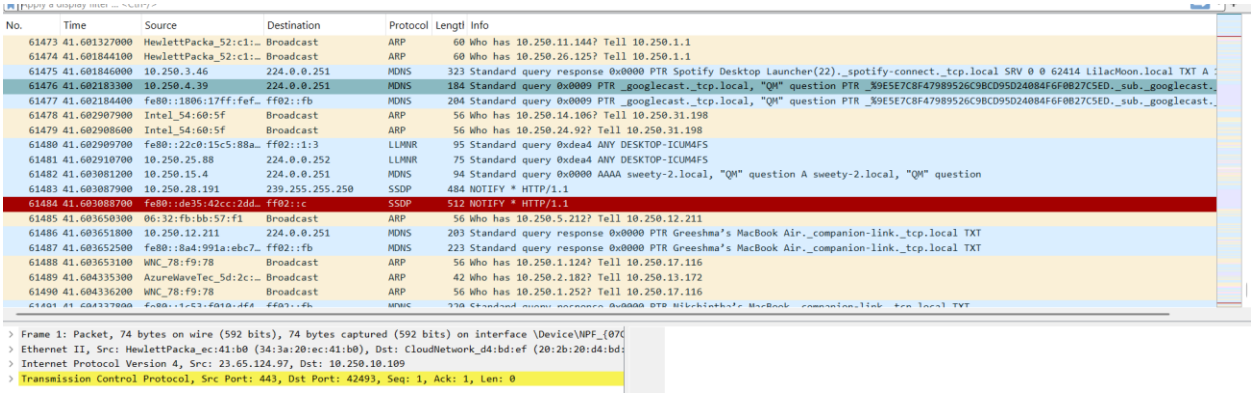
6. What is packet overhead?
Packet overhead is the number of bytes used by protocol headers and other control information rather than by the application payload.

7. How does header overhead affect network efficiency?
Headers use part of the transmitted frame. A larger overhead leaves a smaller proportion for application data, reducing payload efficiency, especially for small packets.

Result

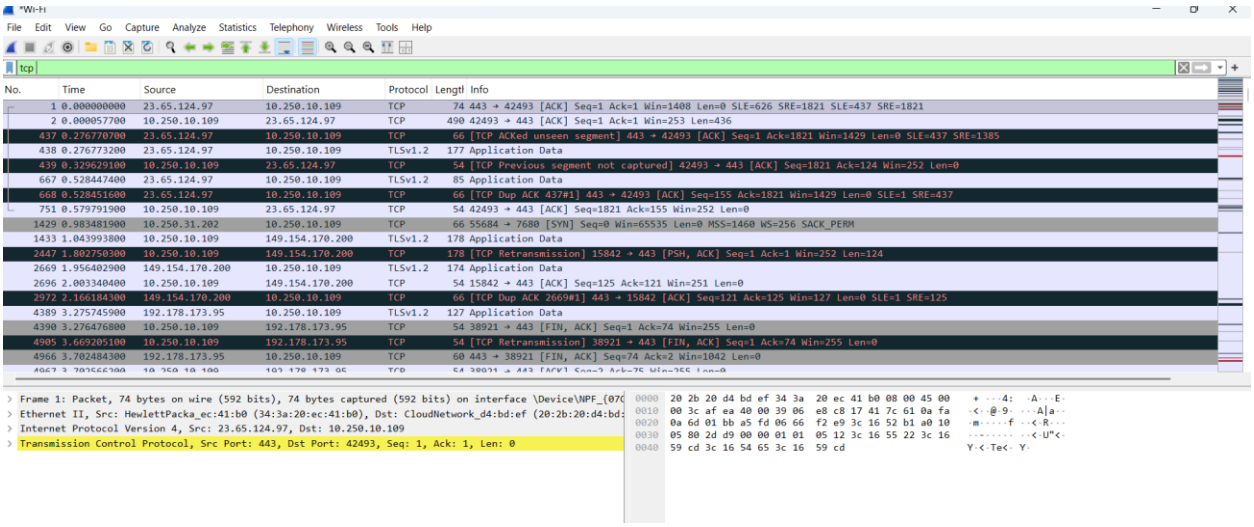
Network packets were captured and examined using Wireshark. Ethernet, IPv4, TCP and UDP header information was identified. For the selected captured packets, the TCP payload efficiency was approximately 88.98%, while the UDP payload efficiency was 60.00%. The observations show that protocol headers contribute to packet overhead and affect the proportion of each frame available for application data.

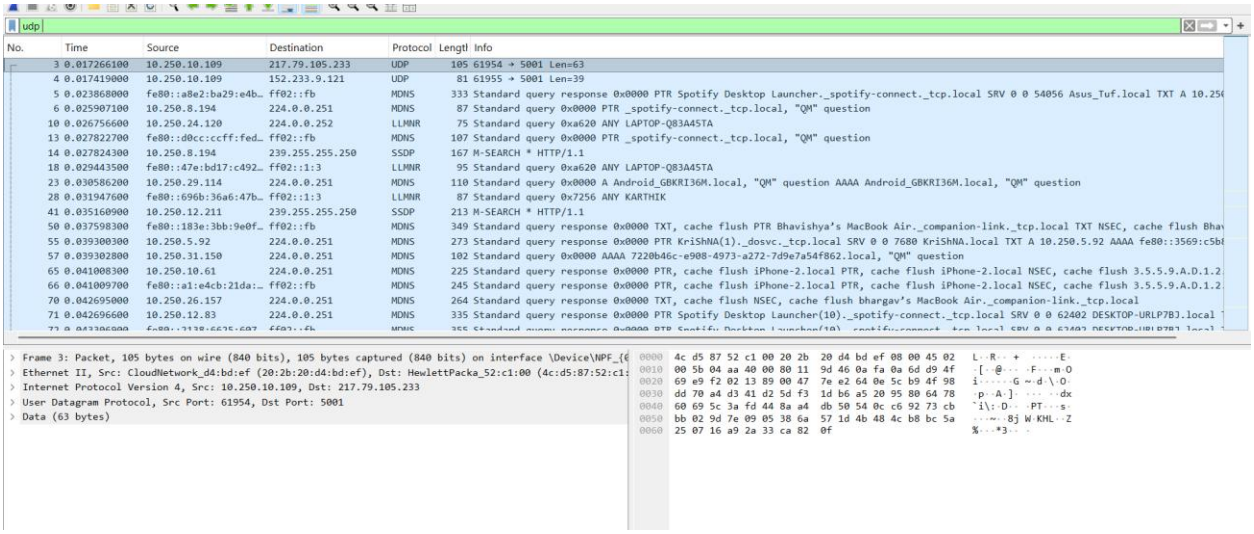
Screenshot Evidence



No.	Time	Source	Destination	Protocol	Lengtl	Info
61473	41.601327000	HewlettPacka_52:c1:…	Broadcast	ARP	60	Who has 10.250.11.144? Tell 10.250.1.1
61474	41.601844100	HewlettPacka_52:c1:…	Broadcast	ARP	60	Who has 10.250.26.125? Tell 10.250.1.1
61475	41.601846000	10.250.3.46	224.0.0.251	MDNS	323	Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 62414 LilacMoon.local TXT A
61476	41.602183300	10.250.4.39	224.0.0.251	MDNS	184	Standard query 0x0009 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast.
61477	41.602184400	fe80::1806:17ff:fef…	ff02::fb	MDNS	204	Standard query 0x0009 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast.
61478	41.602907900	Intel_54:60:5f	Broadcast	ARP	56	Who has 10.250.14.106? Tell 10.250.31.198
61479	41.602908600	Intel_54:60:5f	Broadcast	ARP	56	Who has 10.250.24.92? Tell 10.250.31.198
61480	41.602909700	fe80::22c0:15c5:88a…	ff02::1:3	LLMNR	95	Standard query 0xdea4 ANY DESKTOP-ICUM4FS
61481	41.602910700	10.250.25.88	224.0.0.252	LLMNR	75	Standard query 0xdea4 ANY DESKTOP-ICUM4FS
61482	41.603081200	10.250.15.4	224.0.0.251	MDNS	94	Standard query 0x0000 AAAA sweety-2.local, "QM" question A sweety-2.local, "QM" question
61483	41.603087900	10.250.28.191	239.255.255.250	SSDP	484	NOTIFY * HTTP/1.1
61484	41.603088700	fe80::de35:42cc:2dd…	ff02::c	SSDP	512	NOTIFY * HTTP/1.1
61485	41.603650300	06:32:fb:bb:57:f1	Broadcast	ARP	56	Who has 10.250.5.212? Tell 10.250.12.211
61486	41.603651800	10.250.12.211	224.0.0.251	MDNS	203	Standard query response 0x0000 PTR Greeshma's MacBook Air._companion-link._tcp.local TXT
61487	41.603652500	fe80::8a4:991a:ebc7…	ff02::fb	MDNS	223	Standard query response 0x0000 PTR Greeshma's MacBook Air._companion-link._tcp.local TXT
61488	41.603653100	WNC_78:f9:78	Broadcast	ARP	56	Who has 10.250.1.124? Tell 10.250.17.116
61489	41.604335300	AzureWaveTec_5d:2c:…	Broadcast	ARP	42	Who has 10.250.2.182? Tell 10.250.13.172
61490	41.604336200	WNC_78:f9:78	Broadcast	ARP	56	Who has 10.250.1.252? Tell 10.250.17.116

```
> Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{07C
> Ethernet II, Src: HewlettPacka_ec:41:b0 (34:3a:20:ec:41:b0), Dst: CloudNetwork_d4:bd:ef (20:2b:20:d4:bd:
> Internet Protocol Version 4, Src: 23.65.124.97, Dst: 10.250.10.109
> Transmission Control Protocol, Src Port: 443, Dst Port: 42493, Seq: 1, Ack: 1, Len: 0
```


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 23.65.124.97 10.250.10.109 TCP 74 443 → 42493 [ACK] Seq=1 Ack=1 Win=1408 Len=0 SLE=626 SRE=1821 SLE=437 SRE=1821
2 0.000057700 10.250.10.109 23.65.124.97 TCP 490 42493 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=436
437 0.276770700 23.65.124.97 10.250.10.109 TCP 66 [TCP ACKed unseen segment] 443 → 42493 [ACK] Seq=1 Ack=1821 Win=1429 Len=0 SLE=437 SRE=1385
438 0.276773200 23.65.124.97 10.250.10.109 TLSv1.2 177 Application Data
439 0.329629100 10.250.10.109 23.65.124.97 TCP 54 [TCP Previous segment not captured] 42493 → 443 [ACK] Seq=1821 Ack=124 Win=252 Len=0
667 0.528447400 23.65.124.97 10.250.10.109 TLSv1.2 85 Application Data
668 0.528451600 23.65.124.97 10.250.10.109 TCP 66 [TCP Dup ACK 437#1] 443 → 42493 [ACK] Seq=155 Ack=1821 Win=1429 Len=0 SLE=1 SRE=437
751 0.579791900 10.250.10.109 23.65.124.97 TCP 54 42493 → 443 [ACK] Seq=1821 Ack=155 Win=252 Len=0
1429 0.983481900 10.250.31.202 10.250.10.109 TCP 66 55684 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1433 1.043993800 10.250.10.109 149.154.170.200 TLSv1.2 178 Application Data
2447 1.802750300 10.250.10.109 149.154.170.200 TCP 178 [TCP Retransmission] 15842 → 443 [PSH, ACK] Seq=1 Ack=1 Win=252 Len=124
2669 1.956402900 149.154.170.200 10.250.10.109 TLSv1.2 174 Application Data
2696 2.003340400 10.250.10.109 149.154.170.200 TCP 54 15842 → 443 [ACK] Seq=125 Ack=121 Win=251 Len=0
2972 2.166184300 149.154.170.200 10.250.10.109 TCP 66 [TCP Dup ACK 2669#1] 443 → 15842 [ACK] Seq=121 Ack=125 Win=127 Len=0 SLE=1 SRE=125
4389 3.275745900 192.178.173.95 10.250.10.109 TLSv1.2 127 Application Data
4390 3.276476800 10.250.10.109 192.178.173.95 TCP 54 38921 → 443 [FIN, ACK] Seq=1 Ack=74 Win=255 Len=0
4905 3.669205100 10.250.10.109 192.178.173.95 TCP 54 [TCP Retransmission] 38921 → 443 [FIN, ACK] Seq=1 Ack=74 Win=255 Len=0
4966 3.702484300 192.178.173.95 10.250.10.109 TCP 60 443 → 38921 [FIN, ACK] Seq=74 Ack=2 Win=1042 Len=0
Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{07
Ethernet II, Src: HewlettPacka_ec:41:b0 (34:3a:20:ec:41:b0), Dst: CloudNetwork_d4:bd:ef (20:2b:20:d4:bd:
Internet Protocol Version 4, Src: 23.65.124.97, Dst: 10.250.10.109
Transmission Control Protocol, Src Port: 443, Dst Port: 42493, Seq: 1, Ack: 1, Len: 0
0000 20 2b 20 d4 bd ef 34 3a 20 ec 41 b0 08 00 45 00
0010 00 3c af ea 40 00 39 06 e8 c8 17 41 7c 61 0a fa
0020 0a 6d 01 bb a5 fd 06 66 f2 e9 3c 16 52 b1 a0 10
0030 05 80 2d d9 00 00 01 01 05 12 3c 16 55 22 3c 16
0040 59 cd 3c 16 54 65 3c 16 59 cd

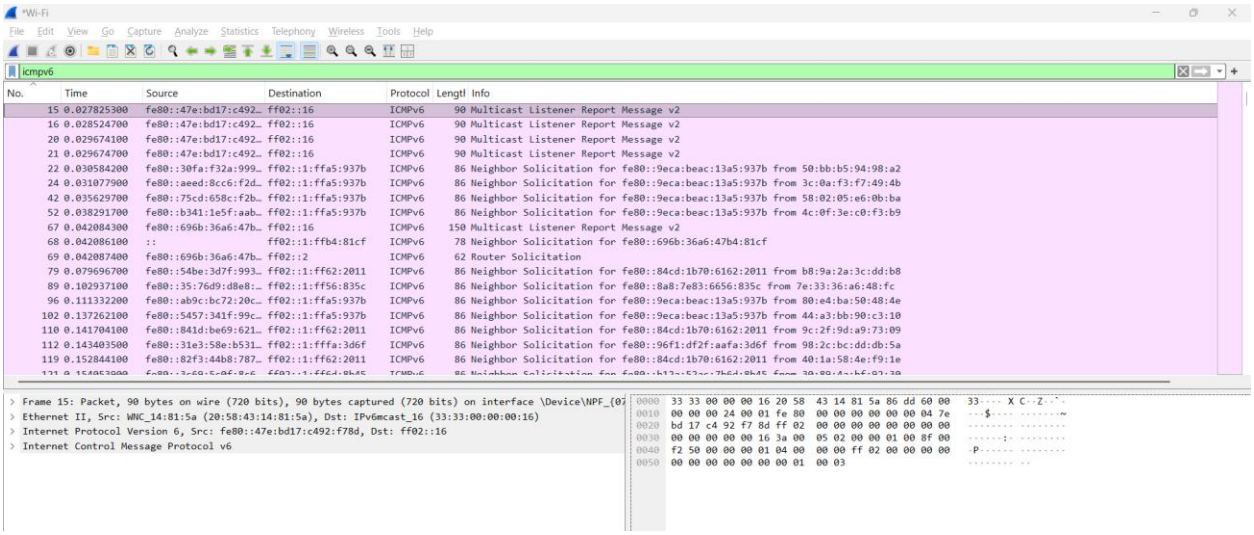


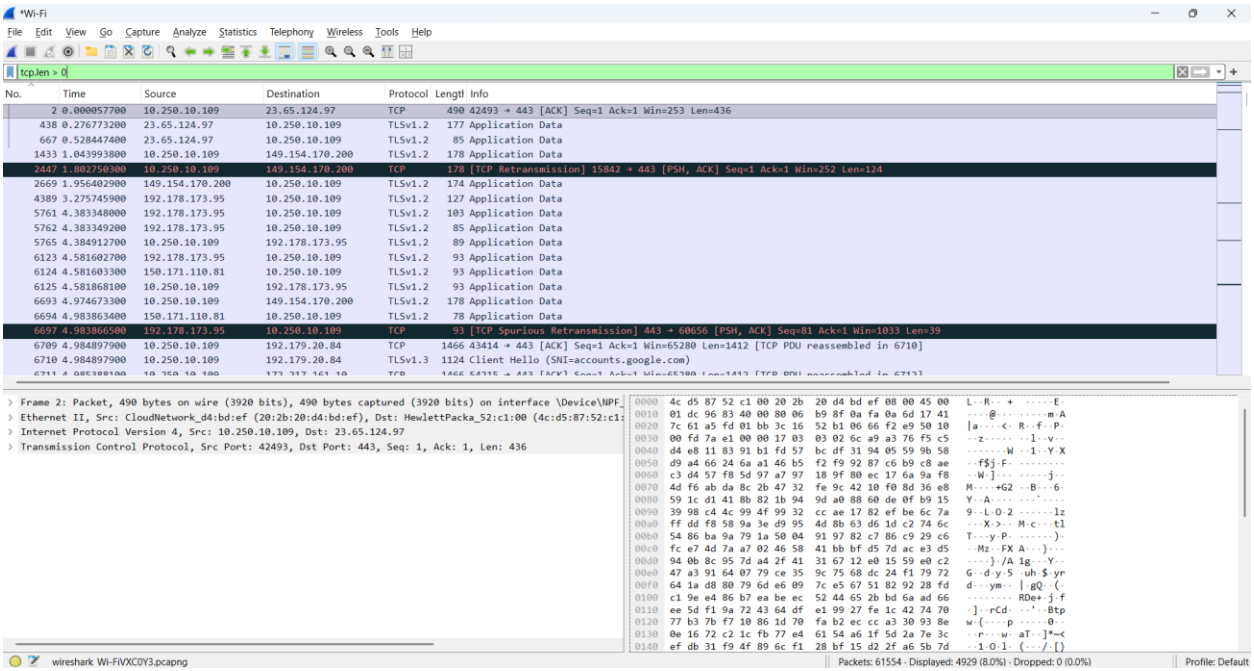
udp

No.	Time	Source	Destination	Protocol	Length	Info
3	0.017266100	10.250.10.109	217.79.105.233	UDP	105	61954 → 5001 Len=63
4	0.017419000	10.250.10.109	152.233.9.121	UDP	81	61955 → 5001 Len=39
5	0.023868000	fe80::a8e2:ba29:e4b…	ff02::fb	MDNS	333	Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 54056 Asus_Tuf.local TXT A 10.25
6	0.025907100	10.250.8.194	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
10	0.026756600	10.250.24.120	224.0.0.252	LLMNR	75	Standard query 0xa620 ANY LAPTOP-Q83A45TA
13	0.027822700	fe80::d0cc:ccff:fed…	ff02::fb	MDNS	107	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
14	0.027824300	10.250.8.194	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
18	0.029443500	fe80::47e:bd17:c492…	ff02::1:3	LLMNR	95	Standard query 0xa620 ANY LAPTOP-Q83A45TA
23	0.030586200	10.250.29.114	224.0.0.251	MDNS	110	Standard query 0x0000 A Android_GBKRI36M.local, "QM" question AAAA Android_GBKRI36M.local, "QM" question
28	0.031947600	fe80::696b:36a6:47b…	ff02::1:3	LLMNR	87	Standard query 0x7256 ANY KARTHIK
41	0.035160900	10.250.12.211	239.255.255.250	SSDP	213	M-SEARCH * HTTP/1.1
50	0.037598300	fe80::183e:3bb:9e0f…	ff02::fb	MDNS	349	Standard query response 0x0000 TXT, cache flush PTR Bhavishya's MacBook Air._companion-link._tcp.local TXT NSEC, cache flush Bha
55	0.039300300	10.250.5.92	224.0.0.251	MDNS	273	Standard query response 0x0000 PTR KriShNA(1)._dosvc._tcp.local SRV 0 0 7680 KriShNA.local TXT A 10.250.5.92 AAAA fe80::3569:c5b
57	0.039302800	10.250.31.150	224.0.0.251	MDNS	102	Standard query 0x0000 AAAA 7220b46c-e908-4973-a272-7d9e7a54f862.local, "QM" question
65	0.041008300	10.250.10.61	224.0.0.251	MDNS	225	Standard query response 0x0000 PTR, cache flush iPhone-2.local PTR, cache flush iPhone-2.local NSEC, cache flush 3.5.5.9.A.D.1.2
66	0.041009700	fe80::a1:e4cb:21da:…	ff02::fb	MDNS	245	Standard query response 0x0000 PTR, cache flush iPhone-2.local PTR, cache flush iPhone-2.local NSEC, cache flush 3.5.5.9.A.D.1.2
70	0.042695000	10.250.26.157	224.0.0.251	MDNS	264	Standard query response 0x0000 TXT, cache flush NSEC, cache flush bhargav's MacBook Air._companion-link._tcp.local
71	0.042696600	10.250.12.83	224.0.0.251	MDNS	335	Standard query response 0x0000 PTR Spotify Desktop Launcher(10)._spotify-connect._tcp.local SRV 0 0 62402 DESKTOP-URLP7BJ.local

```
> Frame 3: Packet, 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on interface \Device\NPF_{6
> Ethernet II, Src: CloudNetwork_d4:bd:ef (20:2b:20:d4:bd:ef), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:
> Internet Protocol Version 4, Src: 10.250.10.109, Dst: 217.79.105.233
> User Datagram Protocol, Src Port: 61954, Dst Port: 5001
> Data (63 bytes)
```

```
0000  4c d5 87 52 c1 00 20 2b  20 d4 bd ef 08 00 45 02   L··R·· +  ·····E·
0010  00 5b 04 aa 40 00 80 11  9d 46 0a fa 0a 6d d9 4f   ·[··@··· ·F···m·O
0020  69 e9 f2 02 13 89 00 47  7e e2 64 0e 5c b9 4f 98   i·····G ~·d·\·O·
0030  dd 70 a4 d3 41 d2 5d f3  1d b6 a5 20 95 80 64 78   ·p··A·]· ··· ··dx
0040  60 69 5c 3a fd 44 8a a4  db 50 54 0c c6 92 73 cb   `i\:·D·· ·PT···s·
0050  bb 02 9d 7e 09 05 38 6a  57 1d 4b 48 4c b8 bc 5a   ···~··8j W·KHL··Z
0060  25 07 16 a9 2a 33 ca 82  0f                       %···*3·· ·
```


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmpv6
No. Time Source Destination Protocol Lengtl Info
15 0.027825300 fe80::47e:bd17:c492… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
16 0.028524700 fe80::47e:bd17:c492… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
20 0.029674100 fe80::47e:bd17:c492… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
21 0.029674700 fe80::47e:bd17:c492… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
22 0.030584200 fe80::30fa:f32a:999… ff02::1:ffa5:937b ICMPv6 86 Neighbor Solicitation for fe80::9eca:beac:13a5:937b from 50:bb:b5:94:98:a2
24 0.031077900 fe80::aeed:8cc6:f2d… ff02::1:ffa5:937b ICMPv6 86 Neighbor Solicitation for fe80::9eca:beac:13a5:937b from 3c:0a:f3:f7:49:4b
42 0.035629700 fe80::75cd:658c:f2b… ff02::1:ffa5:937b ICMPv6 86 Neighbor Solicitation for fe80::9eca:beac:13a5:937b from 58:02:05:e6:0b:ba
52 0.038291700 fe80::b341:1e5f:aab… ff02::1:ffa5:937b ICMPv6 86 Neighbor Solicitation for fe80::9eca:beac:13a5:937b from 4c:0f:3e:c0:f3:b9
67 0.042084300 fe80::696b:36a6:47b… ff02::16 ICMPv6 150 Multicast Listener Report Message v2
68 0.042086100 :: ff02::1:ffb4:81cf ICMPv6 78 Neighbor Solicitation for fe80::696b:36a6:47b4:81cf
69 0.042087400 fe80::696b:36a6:47b… ff02::2 ICMPv6 62 Router Solicitation
79 0.079696700 fe80::54be:3d7f:993… ff02::1:ff62:2011 ICMPv6 86 Neighbor Solicitation for fe80::84cd:1b70:6162:2011 from b8:9a:2a:3c:dd:b8
89 0.102937100 fe80::35:76d9:d8e8:… ff02::1:ff56:835c ICMPv6 86 Neighbor Solicitation for fe80::8a8:7e83:6656:835c from 7e:33:36:a6:48:fc
96 0.111332200 fe80::ab9c:bc72:20c… ff02::1:ffa5:937b ICMPv6 86 Neighbor Solicitation for fe80::9eca:beac:13a5:937b from 80:e4:ba:50:48:4e
102 0.137262100 fe80::5457:341f:99c… ff02::1:ffa5:937b ICMPv6 86 Neighbor Solicitation for fe80::9eca:beac:13a5:937b from 44:a3:bb:90:c3:10
110 0.141704100 fe80::841d:be69:621… ff02::1:ff62:2011 ICMPv6 86 Neighbor Solicitation for fe80::84cd:1b70:6162:2011 from 9c:2f:9d:a9:73:09
112 0.143403500 fe80::31e3:58e:b531… ff02::1:fffa:3d6f ICMPv6 86 Neighbor Solicitation for fe80::96f1:df2f:aafa:3d6f from 98:2c:bc:dd:db:5a
119 0.152844100 fe80::82f3:44b8:787… ff02::1:ff62:2011 ICMPv6 86 Neighbor Solicitation for fe80::84cd:1b70:6162:2011 from 40:1a:58:4e:f9:1e
Frame 15: Packet, 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface \Device\NPF_{07
Ethernet II, Src: WNC_14:81:5a (20:58:43:14:81:5a), Dst: IPv6mcast_16 (33:33:00:00:00:16)
Internet Protocol Version 6, Src: fe80::47e:bd17:c492:f78d, Dst: ff02::16
Internet Control Message Protocol v6
0000 33 33 00 00 00 16 20 58 43 14 81 5a 86 dd 60 00 33····· X C··Z··`·
0010 00 00 00 24 00 01 fe 80 00 00 00 00 00 00 04 7e ···$···· ·······~
0020 bd 17 c4 92 f7 8d ff 02 00 00 00 00 00 00 00 00 ········ ········
0030 00 00 00 00 00 16 3a 00 05 02 00 00 01 00 8f 00 ·····:· ········
0040 f2 50 00 00 00 01 04 00 00 00 ff 02 00 00 00 00 ·P····· ········
0050 00 00 00 00 00 00 00 01 00 03 ········ ··


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len > 0
No. Time Source Destination Protocol Lengtl Info
2 0.000057700 10.250.10.109 23.65.124.97 TCP 490 42493 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=436
438 0.276773200 23.65.124.97 10.250.10.109 TLSv1.2 177 Application Data
667 0.528447400 23.65.124.97 10.250.10.109 TLSv1.2 85 Application Data
1433 1.043993800 10.250.10.109 149.154.170.200 TLSv1.2 178 Application Data
2447 1.802750300 10.250.10.109 149.154.170.200 TCP 178 [TCP Retransmission] 15842 → 443 [PSH, ACK] Seq=1 Ack=1 Win=252 Len=124
2669 1.956402900 149.154.170.200 10.250.10.109 TLSv1.2 174 Application Data
4389 3.275745900 192.178.173.95 10.250.10.109 TLSv1.2 127 Application Data
5761 4.383348000 192.178.173.95 10.250.10.109 TLSv1.2 103 Application Data
5762 4.383349200 192.178.173.95 10.250.10.109 TLSv1.2 85 Application Data
5765 4.384912700 10.250.10.109 192.178.173.95 TLSv1.2 89 Application Data
6123 4.581602700 192.178.173.95 10.250.10.109 TLSv1.2 93 Application Data
6124 4.581603300 150.171.110.81 10.250.10.109 TLSv1.2 93 Application Data
6125 4.581868100 10.250.10.109 192.178.173.95 TLSv1.2 93 Application Data
6693 4.974673300 10.250.10.109 149.154.170.200 TLSv1.2 178 Application Data
6694 4.983863400 150.171.110.81 10.250.10.109 TLSv1.2 78 Application Data
6697 4.983866500 192.178.173.95 10.250.10.109 TCP 93 [TCP Spurious Retransmission] 443 → 60656 [PSH, ACK] Seq=81 Ack=1 Win=1033 Len=39
6709 4.984897900 10.250.10.109 192.179.20.84 TCP 1466 43414 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 6710]
6710 4.984897900 10.250.10.109 192.179.20.84 TLSv1.3 1124 Client Hello (SNI=accounts.google.com)
Frame 2: Packet, 490 bytes on wire (3920 bits), 490 bytes captured (3920 bits) on interface \Device\NPF_
Ethernet II, Src: CloudNetwork_d4:bd:ef (20:2b:20:d4:bd:ef), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:
Internet Protocol Version 4, Src: 10.250.10.109, Dst: 23.65.124.97
Transmission Control Protocol, Src Port: 42493, Dst Port: 443, Seq: 1, Ack: 1, Len: 436
0000 4c d5 87 52 c1 00 20 2b 20 d4 bd ef 08 00 45 00
0010 01 dc 96 83 40 00 80 06 b9 8f 0a fa 0a 6d 17 41
0020 7c 61 a5 fd 01 bb 3c 16 52 b1 06 66 f2 e9 50 10
0030 00 fd 7a e1 00 00 17 03 03 02 6c a9 a3 76 f5 c5
0040 d4 e8 11 83 91 b1 fd 57 bc df 31 94 05 59 9b 58
0050 d9 a4 66 24 6a a1 46 b5 f2 f9 92 87 c6 b9 c8 ae
0060 c3 d4 57 f8 5d 97 a7 97 18 9f 80 ec 17 6a 9a f8
0070 4d f6 ab da 8c 2b 47 32 fe 9c 42 10 f0 8d 36 e8
0080 59 1c d1 41 8b 82 1b 94 9d a0 88 60 de 0f b9 15
0090 39 98 c4 4c 99 4f 99 32 cc ae 17 82 ef be 6c 7a
00a0 ff dd f8 58 9a 3e d9 95 4d 8b 63 d6 1d c2 74 6c
00b0 54 86 ba 9a 79 1a 50 04 91 97 82 c7 86 c9 29 c6
00c0 fc e7 4d 7a a7 02 46 58 41 bb bf d5 7d ac e3 d5
00d0 94 0b 8c 95 7d a4 2f 41 31 67 12 e0 15 59 e0 c2
00e0 47 a3 91 64 07 79 ce 35 9c 75 68 dc 24 f1 79 72
00f0 64 1a d8 80 79 6d e6 09 7c e5 67 51 82 92 28 fd
0100 c1 9e e4 86 b7 ea be ec 52 44 65 2b bd 6a ad 66
0110 ee 5d f1 9a 72 43 64 df e1 99 27 fe 1c 42 74 70
0120 77 b3 7b f7 10 86 1d 70 fa b2 ec cc a3 30 93 8e
0130 0e 16 72 c2 1c fb 77 e4 61 54 a6 1f 5d 2a 7e 3c
0140 ef db 31 f9 4f 89 6c f1 28 bf 15 d2 2f a6 5b 7d
wireshark Wi-FiVXC0Y3.pcapng
Packets: 61554 · Displayed: 4929 (8.0%) · Dropped: 0 (0.0%)
Profile: Default